

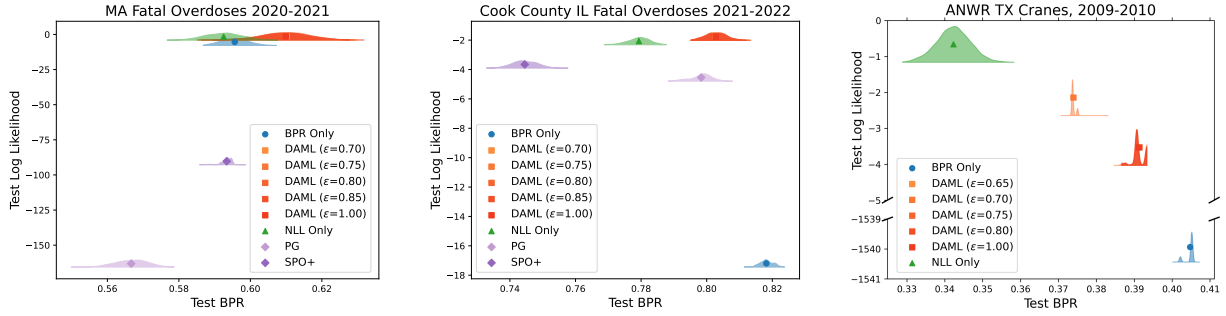


Figure 3: **Pareto frontier of best possible reach (BPR, x-axis) and log likelihood (LL, y-axis) for real-world tasks.** Higher is better on both axes. Each panel how the final models estimated by different training methods score on the test set of a forecasting task defined in Sec. 5. To capture the stochasticity of BPR due to our sampling-based ranking estimator, for each model we show an estimated density for BPR over 1000 trials. Across all 3 tasks, our proposed decision-aware ML (DAML) delivers better top-K decisions as measured by BPR than ML estimation. DAML also delivers likelihood comparable to ML methods and *much better* than directly optimizing BPR.

Additionally, we compare to other decision-aware losses, Perturbed Gradient (PG) and Smart "Predict, then Optimize" (SPO+). For these methods, we use our ratio estimator to rank sites and the gradient of this estimator in 10. We conduct a hyperparameter search over learning rate (and perturbation noise for PG), and select the model with the best loss value on validation data). Unfortunately, these losses lead to numerically unstable solutions for the ANWR TX Cranes dataset, and we were unable to train a model with SPO+ or PG during the response period. We see that these decision-aware losses do indeed produce good top-K decisions, with SPO+ outperforming the ML estimate in Massachusetts and PG doing likewise in Cook County. However, as these models do not train to maximize likelihood, we accordingly get low-likelihood solutions. Furthermore, as they do not directly optimize for the metric of interest, they are outperformed by the DAML models.